



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Quiz 1 – FALL 2025

Course Title:	Digital System Design	Course Code:	CPE 344	Credit Hours:	4(3,
Course Instructor:	Dr. Muhammad Naeem Awais	Program Name:	BS Computer Engineering		
Semester:	7 th	Batch:	FA22	Section:	B
Date:	29 th September, 2025	Maximum Marks:	10		
Student's Name:			Registration Number:		

Question 1: [CLO1-PLO3-C5] [10 Marks]

An immobilizer key is a car key equipped with a transponder chip that wirelessly communicates with the car's security system. When the key is inserted to start the car, the car's onboard computer requests an identifier from the key. If the key transmits the correct identifier, the car starts otherwise the computer disables the ignition system.

Draw a state diagram and design a controller for a Secure Car Key. The system consists of an input a and output of 2-bit identifier. The required system's output transitions are defined as $10 \rightarrow 01 \rightarrow 11 \rightarrow 00$, following the input $a=1$.

